SECTION-D

Note: Long answer type questions. Attempt any one questions out of two questions. (1x10=10)

Q.19 Explain Faraday's Laws of Electromagnetic Induction and their applications.

Q.20 Describe the working and construction of a dynamo, including the force acting on a current carrying conductor in a magnetic field.

Roll No. ....

2nd Sem/ Level-3

DVOC (RAC, MIT, Auto., Servicing, ITM, PT, SD, AMT, FP, EMS)

Subject : Basic Electricity

Time : 2 Hrs. M.M. : 50

SECTION-A

Note: Multiple choice questions. All questions are compulsory (5x1=5)

Q.1 What is the unit of electrical resistance?

a) Volt b) Ohm

c) Watt d) Ampere

Q.2 Which law states that the voltage across a conductor is directly proportional to the current flowing through it?

a) Kirchhoff's law b) Faraday's law

c) Lenz's law d) Ohm's law

Q.3 Which of the following is a primary cell?

a) Lead acid cell

b) Dry cell

c) lithium-ion battery

d) Nickel-cadmium battery

Q.4 What is the main advantages of LED lights over incandescent bulbs?

a) Lower cost

b) Higher power consumption

c) Longer lifespan and energy efficiency

d) Large size

Q.5 What happens when a capacitor is fully charged?

a) It, stops conducting current

b) It discharges automatically

c) It increases voltage indefinitely

d) It becomes a short circuit

SECTION-B

Note: Objective type questions. All questions are compulsory. (5x1=5)

Q.6 What is the relation between electrical power and energy?

Q.7 Define Kirchhoff's Current Law (KCL).

Q.8 Name two applications of an inductor.

Q.9 What is the significance of a form factor in AC circuits?

Q.10 Define mutual inductance.

SECTION-C

Note: Very Short answer type questions. Attempt any six questions out of eight questions. (6x5=30)

Q.11 Explain the difference between AC and DC current.

Q.12 What are the advantages of secondary cells over primary cells?

Q.13 Describe the process of charging and discharging a capacitor.

Q.14 Explain the working principle of an electromagnetic relay.

Q.15 What is the significance of RMS value in AC circuits?

Q.16 Describe the construction and working of an LED.

Q.17 Explain the concept of Q-Factor in RLC circuits.

Q.18 How is electrical power measured in an AC circuit?